Journal of Biomedical Informatics 177 (2026) 105007

Contents lists available at ScienceDirect

Journal of Biomedical Informatics

journal homepage: www.elsevier.com/locate/yjbin

Original Research

CTKGR: temporal knowledge graph enhanced prediction of in-hospital events for pneumonia patients using electronic health records

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ARTICLE INFO

Keywords:
Temporal knowledge graph
Clinical event prediction
Electronic health records
Graph convolutional network
Dynamic clinical decision support

ABSTRACT

Objective: This study aims to develop an advanced clinical event prediction model leveraging the temporal characteristics embedded within electronic health record (EHR), with a specific focus on predicting clinical events during the hospitalization of pneumonia patients to enhance clinical decision-making.
Methods: We constructed a temporal knowledge graph (TKG) by systematically extracting clinical information from pneumonia patient EHR data across multiple time points. To effectively capture temporal dynamics and complex clinical interactions, we introduced the clinical TKG representation (CTKGR) model. This innovative model integrates graph convolutional networks (GCN), recurrent neural networks (RNN), and attention mechanisms to explicitly model clinical event progression. Additionally, a time length-aware module (TLAM) was developed to address challenges posed by irregular time intervals and varying frequencies of patient visits.
Results: The proposed CTKGR model exhibited superior predictive performance, achieving an AUC of 0.745 for symptom prediction and 0.798 for medication recommendation tasks, significantly outperforming traditional static and non-temporal methods.
Conclusion: The CTKGR model effectively captures the temporal dynamics inherent in clinical data, demonstrating robust predictive capabilities for dynamically evolving clinical events. This study offers a novel and effective framework for AI-driven clinical decision support, with strong potential for application in pneumonia management.

Problem	Existing clinical decision support systems primarily rely on static patient data from Electronic Health Records (EHRs), inadequately leveraging temporal dynamic information. This limitation hinders accurate modeling of disease progression and personalized clinical event prediction, especially evident in diseases like pneumonia with highly dynamic clinical courses.
What is Already Known	1. Deep learning methods, including RNN, LSTM, and Transformer models, have shown potential in modeling sequential clinical data, effectively capturing temporal patterns in medical tasks. 2. Knowledge Graphs (KGs) provide structured representations of medical knowledge and have been successfully applied in clinical predictions, but traditional KGs typically neglect temporal dimensions.

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Problem	Existing clinical decision support systems primarily rely on static patient data from Electronic Health Records (EHRs), inadequately leveraging temporal dynamic information. This limitation hinders accurate modeling of disease progression and personalized clinical event prediction, especially evident in diseases like pneumonia with highly dynamic clinical courses.
What is Already Known	3. Temporal Knowledge Graphs (TKGs) have been effectively applied in other domains (finance, social networks) but face challenges in medical scenarios due to irregular intervals, and event heterogeneity.
What This Paper Adds	This study introduces the Clinical Temporal Knowledge Graph Representation (CTKGR) model, combining Graph Convolutional Network (GCN), RNN, and attention

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https://doi.org/10.1016/j.jbi.2026.105007
Received 7 August 2025; Received in revised form 6 January 2026; Accepted 2 March 2026
Available online 4 March 2026
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Problem	Existing clinical decision support systems primarily rely on static patient data from Electronic Health Records (EHRs), inadequately leveraging temporal dynamic information. This limitation hinders accurate modeling of disease progression and personalized clinical event prediction, especially evident in diseases like pneumonia with highly dynamic clinical courses.
	mechanisms to explicitly model temporal clinical dynamics. A novel Time Length-Aware Module (TLAM) is developed within the CTKGR framework to address practical challenges such as irregular time intervals and inconsistent patient visit frequencies.

1. Introduction

With the continuous advancement of healthcare information, Electronic Health Records (EHRs) have emerged as essential repositories for comprehensive patient data, playing pivotal roles in disease risk assessment, treatment outcome prediction, prognosis evaluation, and healthcare resource allocation optimization [1–3]. Pneumonia, a prevalent and severe infectious disease characterized by high incidence and pathogen diversity, poses substantial threats to public health. EHRs of pneumonia patients contain detailed clinical data, including medical history, symptoms, diagnostic test results, treatments, and medications, providing valuable insights into pneumonia pathogenesis, clinical characteristics, and therapeutic strategies [4,5].

The onset and progression of pneumonia demonstrate distinct dynamic patterns, requiring clinicians to consider daily variations in patient conditions and treatment adjustments during hospitalization. However, current clinical decision support systems primarily rely on static data features, insufficiently utilizing the temporal information contained within EHRs. Consequently, they often fail to accurately capture disease progression trends and individualized treatment requirements. Therefore, effectively extracting temporal dynamic features from medical data and precisely predicting clinical events remain critical challenges in medical big data analytics and AI (artificial intelligence) research [6–8].

In recent years, temporal models based on deep learning have made substantial progress in medical prediction tasks. RNN (Recurrent Neural Network) and their variants, such as LSTM (Long Short-Term Memory), have been extensively applied to longitudinal EHR analyses due to their strong capability in sequential modeling [9]. For instance, Liu et al. proposed the KIT-LSTM model, integrating temporal and knowledge-aware mechanisms, significantly improving the accuracy and interpretability of continuous mortality risk prediction [10]. Additionally, Transformer-based models have gained increasing attention due to their superior ability to capture long-range dependencies in medical time series data [11,12]. Chen et al. introduced the TSOANet model employing an orthogonal attention mechanism to represent event types and temporal intervals effectively, achieving good performance in modeling disease progression [13]. Nevertheless, these methods still face challenges in addressing practical issues, such as diverse event types, irregular intervals, and multi-task clinical predictions, thus inadequately capturing the complexities of real-world clinical scenarios.

KG (Knowledge Graph), characterized by structured knowledge representation, have been widely used in medical knowledge modeling and disease prediction tasks [14–17]. However, traditional KGs generally neglect the temporal dimension, limiting their effectiveness in capturing the dynamic progression of medical events. Recently, TKG (Temporal Knowledge Graph) has gained popularity [18,19], with notable success in fields such as finance and social networks. Nonetheless, their application in the medical domain remains challenging due to issues like variable-length patient records, difficulties in aligning temporal data across individuals, and event heterogeneity.

To address these challenges, this study proposes the CTKGR (Clinical Temporal Knowledge Graph Representation) model, focusing on the hospitalization course of pneumonia patients. The CTKGR model combines GCN (Graph Convolutional Network), RNN (Recurrent Neural Network), and attention mechanisms to effectively model clinical temporal dynamics. Additionally, the innovative TLAM (Time Length-Aware Module) is designed within the CTKGR framework to effectively manage irregular time intervals and data missingness. Experimental validations conducted across several clinical tasks, including symptom prediction and medication recommendation, demonstrate the effectiveness of the CTKGR model. This research significantly advances dynamic clinical event prediction and provides valuable theoretical and practical insights for the broader application of TKGs in healthcare.

2. Method

2.1. Dataset

The dataset in this study consists of medical records from 3,214 patients diagnosed with community-acquired pneumonia admitted to the respiratory department of a tertiary hospital in Beijing from January 2019 to June 2022. The exclusion criteria included patients admitted due to COVID-19 during 2021–2022 and patients with concomitant severe illnesses, such as cancer, cardiovascular diseases, or severe liver and kidney diseases. Ethical approval for data collection and use was granted by the hospital's ethics committee (Approval Number: 2023-P2-031–01).

This study involved patients diagnosed with community-acquired pneumonia who were admitted to a tertiary hospital in Beijing between January 2019 and June 2022. To ensure the reliability and integrity of the data, the following inclusion and exclusion criteria were applied to extract eligible EHRs:

1) Only patients with a confirmed diagnosis of pneumonia made by attending physicians during hospitalization were included (ICD-10 J12–J18); patients hospitalized due to COVID-19 during 2021–2022 were excluded.
2) For patients with multiple hospitalizations, only the first admission record was retained for analysis, although subsequent readmissions were labeled accordingly.
3) Patients with conditions likely to dominate the clinical course and fundamentally alter care pathways, such as end-of-life/comfort care at admission or profound immunosuppression were excluded.
4) Patients who were reported as deceased in their discharge records were also excluded.

The medical records encompass three critical stages: admission, hospitalization course, and discharge. Admission records include personal information, chief complaints, current medical history, past medical history, personal history, marital and reproductive history, family history, auxiliary examinations, and admission diagnoses. Hospitalization records consist of progressive diagnoses, diagnostic rationale, differential diagnoses, treatment plans, disease progression, and physician round notes. Discharge records contain admission conditions, progressive diagnoses, treatment processes, discharge diagnoses, and discharge instructions. The details of different types of data were shown in Supplementary Table 1.

2.2. Construction of TKG for pneumonia disease progression

To effectively describe and analyze disease progression, this study constructed a temporal knowledge graph of pneumonia disease progression from medical records, as illustrated in Fig. 1. This graph integrates medical concepts, relationships, and temporal information at each stage of disease progression, supplemented and optimized with multi-source external knowledge.

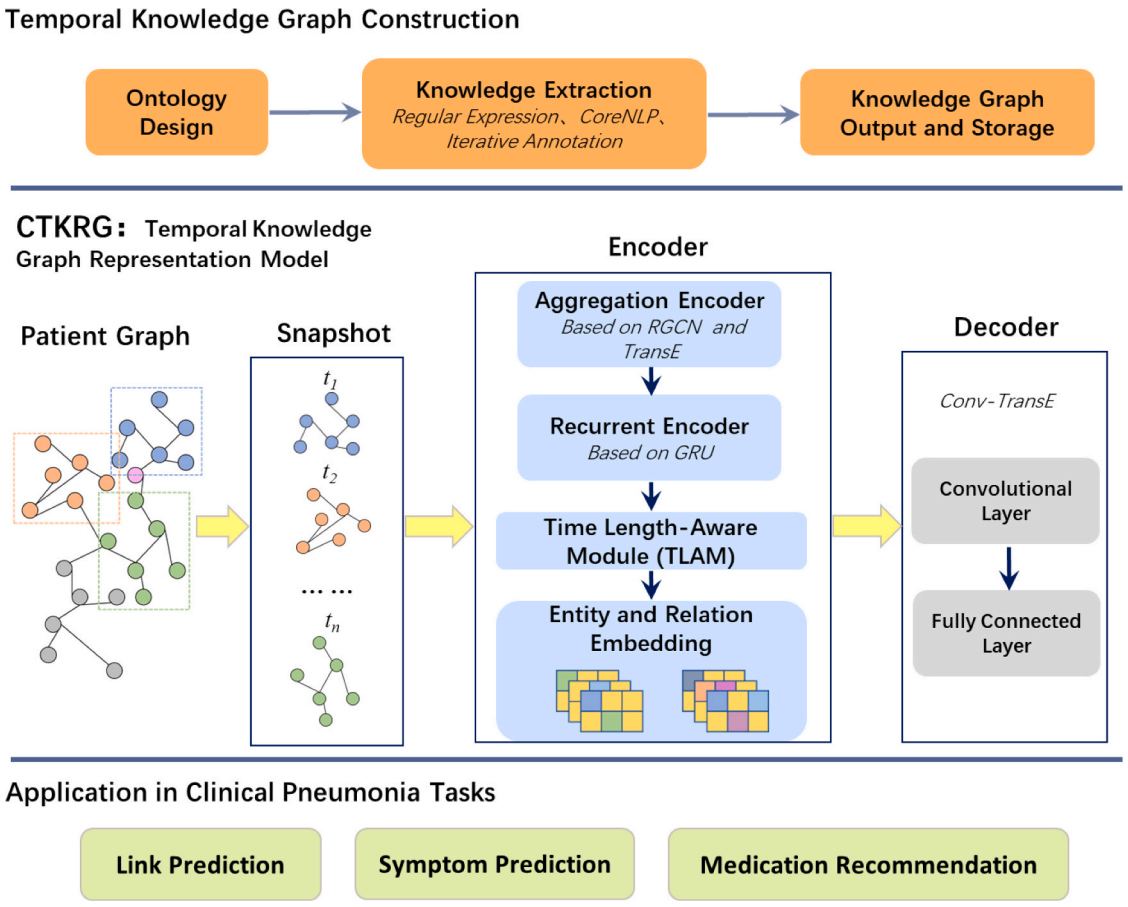


Fig. 1. Workflow of TKG enhanced prediction of in-hospital events for pneumonia.

1) Ontology Design

Our ontology and knowledge graph were developed using a five-step construction framework designed for modularity:

- Concept definition: We referred to the design principles of the Open Medical and Healthcare Alliance (OMAHA)[20] and SNOMED-CT [21] ontologies to structure pneumonia-related information and define the key clinical concepts used in this study. The OMAHA schema provides a general-purpose framework for medical knowledge modeling, and, together with SNOMED CT, served as a conceptual reference to guide the development of our domain-specific pneumonia knowledge graph.
- Class hierarchy construction: For each concept, we established a hierarchical structure with multiple subclasses, as shown in Supplementary Table 2.
- Attribute definition: We summarized the characteristics and properties of each class to ensure semantic consistency.
- Relationship specification: We defined the semantic relationships among the classes, as listed in Table 1.
- Data layer integration: We extracted entities and relations from clinical records to populate the data layer and integrated it with the ontology layer to form the complete knowledge graph.

2) Knowledge Extraction

In this study, we first applied regular expression to segment EHRs according to physician-provided templates. Structured data were directly extracted and stored, while unstructured free-text data were processed using the CoreNLP[22] toolkit for initial entity recognition and relation identification. Subsequently, 100 medical records were manually corrected and annotated to create the first-round training corpus. This corpus was used to train a BERT-CRF model for named entity recognition and relation extraction. The trained model was then applied to another 100 medical records for automatic annotation. The results were manually reviewed and corrected, then combined with the previous corpus for the next training cycle. This iterative annotation, training, and correction process continued until all records were fully annotated. Entity standardization was conducted based on ontology classification, and a specialized pneumonia vocabulary was developed, including CMeSH, ICD-10, and expert consultations. The final knowledge graph comprised 5,231 nodes and 94,730 relationships, stored in the Neo4j database in the format of quadruples (entity, relationship, entity, time).

Table 1
Relationships of different class in ontology.

Class	Relation	Relation number
Patient-disease	Has	1
Disease-treatments	Surgical treatment	2
	Pharmacological treatment	3
Disease-medications	Medication treatment	4
Disease-examinations	Lab tests	5
	Imaging examinations	6
	Auxiliary examinations	7
Body parts-symptoms	Has	8
Disease-symptoms	Has	9
Pharmacological treatment-medications	Anti-infective drugs	10
	Respiratory agents	11
	Antiallergic agents	12
	Adjunctive therapies	13

2.3. Pneumonia TKG representation

In contrast to general-domain temporal knowledge graphs, clinical temporal knowledge graphs are characterized by event sequences derived from patients' hospitalization records, where timestamps correspond to clinical documentation times. These graphs present unique challenges, such as patient-specific sequences of varying lengths and irregular time intervals. To address these issues, we proposed a personalized representation framework for patient-level TKG.

We designed a CTKGR model based on the Relational Graph Convolutional Network (RGCN) architecture. The model is specifically tailored to capture the dynamic and heterogeneous nature of clinical temporal graphs. CTKGR jointly models both the structural evolution of the patient-specific graph and the temporal progression of clinical events. The overall architecture of the proposed model is illustrated in Fig. 1.

In the overall representation of TKG, the CTKGR framework adopts a probabilistic modeling paradigm. Given a patient-specific graph denoted as $G_p = (E_p, R_p, T_p)$, the graph includes all entities and relations related to the patient node within g. According to the timestamps recorded during the patient's hospitalization, the patient graph is divided into a sequence of temporal graph snapshots $G_p = \{\widetilde{G}_1, \widetilde{G}_2, \cdots, \widetilde{G}_{|l|}\}$, where $\widetilde{G}_\tau$ denotes the subgraph at time τ, and $|l|$ represents the total number of timestamps in the patient's hospitalization records. Analogous to text sequence representation in language models, the knowledge representation model aims to fit the joint distribution function $P(\widetilde{G}_1, \widetilde{G}_2, \cdots, \widetilde{G}_{|l|})$ of the subgraph sequence. Based on the conditional probability formula, it can be represented as follows:

$$P\left(\widetilde{G}_1, \widetilde{G}_2, \cdots, \widetilde{G}_{|l|}\right) = \prod_{t=1}^{|l|} P\left(\widetilde{G}_t \middle| \widetilde{G}_1, \widetilde{G}_2, \cdots, \widetilde{G}_{t-1,}\right)$$

Subsequently, $\{\widetilde{G}_1, \widetilde{G}_2, \cdots, \widetilde{G}_{t-1,}\}$ is treated as a latent contextual representation, which is approximated using autoregressive models such as RNN. The CTKGR architecture is composed of two primary components: an encoder, which captures the structural and temporal context of the patient-specific graph, and a decoder, which models the conditional generation of future subgraphs based on historical representations.

1) Encoder

A. Aggregation Encoder

Each subgraph contains a sequence of entities $(\widetilde{E}_1, \widetilde{E}_2, \cdots, \widetilde{E}_{|l|})$ and a sequence of relationships $(\widetilde{R}_1, \widetilde{R}_2, \cdots, \widetilde{R}_{|l|})$. For each subgraph, CTKGR models the entity set $\widetilde{E}$ and the relation set $\widetilde{R}$ by first capturing inter-entity associations through factual triples. Furthermore, it explores higher-order relational dependencies by leveraging shared entities as bridges to uncover implicit connections among relations. During model training, the embedding representations of entities and relationships are optimized through the entity aggregation process. Considering the sensitivity of the RGCN network to the number of relations, CTKGR integrates RGCN with the TransE model into a unified aggregation encoder to encode embeddings for entities and relationships. Specifically, CTKGR maintains two matrices $V_{\widetilde{E}}$ and $V_{\widetilde{R}}$ to store entity vectors and relation vectors, respectively. They are initialized randomly and updated during training to capture semantic and structural information in the temporal knowledge graph.

Subsequently, CTKGR updates these matrices through aggregation operations involving the current node and its neighbor nodes under different relationships. This aggregation process is detailed in the following equation:

$$v_{e_i}^{(l+1)} = \sigma\left(\sum_{r\in\widetilde{R}}\sum_{j\in\mathscr{N}_{e_i}^{r}} \frac{1}{\left|\mathscr{N}_{e_j}^{r}\right|} W_1^{(l)}\left(v_{e_j}^{(l)} + v_r\right) + W_0^{(l)} v_{e_i}^{(l)}\right)$$

Here, $v_{e_i}^{(l)}$ represents the vector embedding of node e_i at layer l, and σ denotes the softmax activation function. $\mathscr{N}_{e_i}^{r}$ denotes the set of all neighboring nodes connected to node e_i via relation r, used to normalize the vector outputs. $V_r^{(l)}$ represents the feature matrix of relation r. $W_1^{(l)}$ is a parameter matrix of a fully connected layer without bias, used to map the translation of node e_i under its relation, while $W_0^{(l)}$ is employed for self-loop mapping of node e_i. Through iterative updates using the multi-layer aggregation encoder, embedding representations of entities and relationships in each subgraph are obtained.

B. Recurrent encoder

The entity embedding $v_{e_i,t}$ at the current time is influenced not only by the current related relationship embedding $v_{r,t}$, but also by its embedding at the previous time $v_{e_i,t-1}$. Similarly, relationship embeddings follow the same principle. CTKGR employs two separate GRUs with distinct parameters to capture sequential evolution for entities and relationships, respectively. Considering the correlation between entities and relationships, CTKGR performs average pooling between the current relationship (entity) and its associated entities (relationships) from the previous timestamp. The resulting aggregated representation is then concatenated with the current embedding of the target entity or relation, and the combined vector is subsequently fed into a GRU to capture temporal dependencies. The evolutionary patterns of entities and relationships are represented by the following equations:

$$v_{e_i,t} = GRU\left(v_{e_i,t-1}, \left[f\left(v_{r,t-1}, \mathscr{G}_{e_i}^{r}\right); v_{e_i,t}\right]\right)$$

$$v_{r_i,t} = GRU\left(v_{r_i,t-1}, \left[f\left(v_{e,t-1}, \mathscr{G}_{r_i}^{e}\right); v_{r_i,t}\right]\right)$$

Here, f denotes the average pooling function, $\mathscr{G}_{e_i}^{r}$ represents all relationships associated with e_i, and $\mathscr{G}_{r_i}^{e}$ denotes all entities associated with r_i.

C. TLAM

Due to the unique characteristics of clinical visits, patient records are often recorded at irregular intervals, leading to issues such as missing data and variable-length data sequences. To address these challenges, CTKGR incorporates TLAM, which is an attention-based and time interval-based hidden state interpolation method. TLAM compensates for data gaps by interpolating the hidden states of GRU units at missing timestamps. The standard hidden state $\widetilde{h}_t$ of the GRU unit is calculated according to the following equation:

$$\widetilde{h}_t = tanh(W\cdot[r_t * h_{t-1}, x_t])$$

Here, W represents the trainable parameters, r_τ is the output of the GRU reset gate, h_{t-1} denotes the hidden state output from the previous timestep, and x_t represents the input at the current timestep. An indicator matrix l is introduced to denote missing timestamps. If the input at timestep t is missing, attention scores are computed using the hidden state from timestep t and the hidden states from all intermediate timesteps. Additionally, to incorporate the effect of time intervals, the method provided in ATTAIN [23] is utilized to calculate the time interval scores.

$$s(\Delta t) = \frac{1}{log(e + \Delta t)}$$

Subsequently, the hidden states from all previous timestamps are weighted and summed according to their time interval and attention scores to obtain the interpolated vector for the missing timestamp.

2) Decoder

The decoder for the KG, also known as the scoring function, aims to evaluate the probability of correctness of a given triple. It optimizes an objective function to rationalize the representations of entities and relations. In CTKGR, Conv-TransE is utilized as the decoder, composed of a one-dimensional convolutional layer followed by a fully connected layer. The scoring function to assess the validity of a specific entity-relation pair is calculated using the following equation:

$$s(e_1|e_2, r, \mathrm{t}) = \sigma\left(ConvTransE\left(v_{e_1,\mathrm{t}}, v_{r,\mathrm{t}}\right)\right)$$

$$s(r|e_1, e_2, t) = \sigma\left(ConvTransE\left(v_{e_1,t}, v_{e_2,t}\right)\right)$$

The link prediction tasks targeting entities $(?, e_2, r, t)$ and relations $(e_1, e_2, ?, t)$ can both be formulated as multi-label classification tasks, where each class corresponds to an entity or relation. The learning objective is to minimize the cross-entropy loss function across all quadruple events:

$$\mathscr{L}_e = -\sum_{t\in T}\sum_{i\in E}\sum_{j=1}^{|E|} y_{j,t+1} log(s(e_1|e_2, r, t))$$

$$\mathscr{L}_r = -\sum_{t\in T}\sum_{i\in E}\sum_{j=1}^{|R|} y_{j,t+1} \log(s(r|e_1, e_2, t))$$

where $y_{j,t+1}$ denotes the ground-truth label of the quadruple. Finally, CTKGR is jointly trained by combining these two loss functions with weighted coefficients.

2.4. Model training and evaluation

During the processing of hospitalization data for pneumonia patients, we discretized the length of hospitalization to simplify the modeling task and enhance the generalization capability of the model. The specific discretization rules were as follows: hospital stays of 5 days or fewer were categorized as Class 1; stays between 6 and 14 days were divided into Classes 2 to 10; and stays exceeding 14 days were categorized as Class 11. The Distribution of length of hospital stay was presented in Supplementary Fig. 1.

All deep learning models were developed based on the PyTorch framework and trained on an RTX 3090 GPU. These configurations were optimized through experiments to ensure optimal model performance.

This study evaluated model performance from three perspectives: (1) Link Prediction Task; (2) Medical Event Prediction Task during pneumonia hospitalization; (3) Ablation Study.

To comprehensively evaluate the modeling capability of CTKGR in KG representation learning, we compared it against current mainstream static and temporal KG inference models. The baseline models included three static graph models (TransE, DisMult, RotatE [24]) and four temporal graph models (TTransE [25], TA-DisMult [26], xERTE [27], RE-GCN [28]). Experiments were conducted on two datasets: the self-constructed pneumonia TKG and the public ICEWS18 dataset. The ICEWS18 dataset comprises international political event data from January to October 2018, including 23,033 entities and 256 relation types. Both datasets were temporally partitioned into training, validation, and test sets with a ratio of 8:1:1 to ensure the model's robustness in temporal modeling and generalization.

In the link prediction task, entities or relations in the knowledge graph were randomly masked, and the model predicted the missing parts. Evaluation metrics included Hit@K and Mean Reciprocal Rank (MRR). Hit@K measures the model's ability to correctly predict the entity or relation within the top-K candidate results, while MRR assesses the overall ranking quality of all predictions. Commonly used K-values are 1, 3, and 10, addressing both precision and recall requirements.

In the medical event prediction task during pneumonia hospitalization, we further assessed the performance of CTKGR across multiple clinical tasks and systematically compared it with mainstream models such as Logistic Regression(LR), Random Forest (RF), LSTM, Time-aware LSTM (TLSTM), ATTAIN, VHP, GRU, FLTD[29], Transformer, and StageNet[30]. This task encompasses medication recommendation and symptom prediction, both considered entity-linking prediction tasks. The evaluation metrics used to assess the inference capability of models on relationships between clinical entities included Accuracy (ACC), Area Under the Receiver Operating Characteristic Curve (AUROC), and Area Under the Precision-Recall Curve (AUPRC). ACC measures overall classification correctness, AUROC evaluates model robustness across different decision thresholds, and AUPRC is suitable for assessing model performance on minority classes when facing class imbalance.

3. Results

3.1. Link prediction task

The model performance results for the link prediction task are presented in Table 2. Overall, CTKGR achieved the best performance across the three critical metrics—Hits@1, Hits@3, and Hits@10—improving upon the best baseline model by 3.76%, 7.75%, and 4.68%, respectively.

Further link prediction experiments conducted on the Pneumonia Temporal Knowledge Graph are shown in Table 3. CTKGR consistently achieved optimal performance across all metrics, further validating its effectiveness and generalizability in handling irregular medical temporal data.

3.2. Medical event prediction task during pneumonia hospitalization

In the medical tasks of symptom prediction and medication recommendation, we compared the performance of the CTKGR model with multiple baseline models, with results detailed in Table 4. CTKGR consistently outperformed the selected baseline models across both tasks. Models without temporal information (LR and RF) achieved the lowest performance. Time-aware models such as T-LSTM and ATTAIN outperformed the basic sequential models (GRU and LSTM), confirming the importance of incorporating temporal intervals in modeling medical events. However, in these tasks characterized by imbalanced sample distributions, CTKGR exhibited relatively weaker performance in the AUPRC metric compared to some baseline models.

3.3. Ablation study

We conducted an ablation study to further investigate the contributions of the TLAM module and its variants within the CTKGR model. Three model variants were specifically designed for comparison:

- GRU model: Missing timestamps were filled with nearest-neighbor clinical texts. Each timestamp's clinical records were processed using GRU networks, and representations from all timestamps were concatenated. This method explicitly addresses missing data filling but overlooks interactions between GRU hidden states and temporal interval factors.
- CTKGR-TLAM-Att model: Modified from the CTKGR's TLAM module, this variant considers only time interval factors to model interactions among GRU hidden states preceding missing timestamps, neglecting attention mechanism interactions, potentially limiting its capability in implicitly filling missing data.
- CTKGR-TLAM-TI model: In contrast to CTKGR-TLAM-Att, this variant employs only the attention mechanism to capture dynamic relationships between GRU hidden states, excluding time interval influences.

Table 2
Link Prediction Results on the ICEWS18 Dataset (Mean with 95% Bootstrap Confidence Intervals, 100 Replicates).

	Models	Hits@1	Hits@3	Hits@10	MRR
Static Graph Representation Models	TransE	2.87 (2.55–3.20)	26.87 (26.52–27.28)	43.91 (43.66–44.15)	17.27 (17.04–17.59)
	DisMult	11.36 (10.39–12.28)	25.94 (25.56–26.31)	41.96 (41.60–42.24)	21.88 (20.90–23.15)
	RotatE	14.31 (14.07–14.58)	27.26 (26.94–27.55)	38.46 (37.16–39.51)	22.85 (21.95–23.66)
Temporal Graph Representation Models	TTransE	2.32 (0.53–4.89)	7.94 (7.43–8.48)	21.87 (21.72–22.03)	8.46 (8.35–8.56)
	TA-DisMult	20.19 (20.08–20.30)	31.52 (30.53–32.43)	45.47 (45.10–45.89)	28.42 (27.64–29.41)
	RE-GCN	25.03 (24.71–25.37)	35.58 (35.37–35.75)	47.01 (46.17–48.23)	32.75 (32.50–32.99)
	xERTE	32.47 (31.27–33.76)	40.37 (40.29–40.44)	49.81 (49.36–50.30)	37.19 (36.44–37.81)
	CTKGR	33.69 (33.35–34.09)	43.50 (43.00–44.01)	52.14 (52.04–52.25)	34.78 (34.50–34.99)

Table 3
Link Prediction Results on the Pneumonia TKG (Mean with 95% Bootstrap Confidence Intervals, 100 Replicates).

	Models	Hits@1	Hits@3	Hits@10	MRR
Static Graph Representation Models	TransE	5.61 (5.33–5.88)	15.34 (14.21–16.16)	24.31 (23.71–24.89)	13.16 (12.44–14.15)
	DisMult	13.56 (11.66–15.09)	22.36 (21.94–22.75)	29.47 (28.96–30.04)	20.73 (19.98–21.43)
	RotatE	18.23 (18.04–18.42)	24.15 (23.95–24.34)	34.25 (33.56–34.98)	22.01 (21.51–22.50)
Temporal Graph Representation Models	TTransE	12.48 (12.39–12.56)	16.82 (16.48–17.21)	39.46 (39.26–39.69)	16.15 (16.03–16.27)
	TA-DisMult	24.63 (24.28–24.96)	33.24 (33.16–33.32)	41.65 (41.15–42.14)	30.41 (30.28–30.56)
	CTKGR	37.24 (37.11–37.38)	42.86 (42.75–42.97)	57.17 (56.96–57.37)	40.53 (40.24–40.85)

Table 4
Symptom Prediction Results (Mean with 95% Bootstrap Confidence Intervals, 100 Replicates).

Models	ACC	AUROC	AUPRC
LR	0.582 (0.574–0.591)	0.582 (0.566–0.597)	0.465 (0.451–0.479)
RF	0.597 (0.590–0.604)	0.570 (0.560–0.580)	0.442 (0.433–0.451)
LSTM	0.734 (0.731–0.736)	0.674 (0.674–0.675)	0.535 (0.532–0.538)
VHP	0.773 (0.763–0.783)	0.664 (0.662–0.666)	0.574 (0.572–0.576)
GRU	0.760 (0.755–0.765)	0.687 (0.683–0.692)	0.617 (0.614–0.620)
TLSTM	0.748 (0.744–0.753)	0.707 (0.704–0.710)	0.606 (0.604–0.608)
ATTAIN	0.777 (0.776–0.778)	0.730 (0.723–0.738)	0.622 (0.622–0.623)
FLTD	0.778 (0.778–0.779)	0.706 (0.704–0.707)	0.620 (0.618–0.621)
Transformer	0.784 (0.781–0.786)	0.722 (0.720–0.724)	0.637 (0.633–0.640)
StageNet	0.774 (0.770–0.778)	0.719 (0.716–0.722)	0.624 (0.619–0.629)
CTKGR	0.806 (0.804–0.807)	0.745 (0.744–0.746)	0.427 (0.424–0.430)

Table 5
Medication Recommendation Results (Mean with 95% Bootstrap Confidence Intervals, 100 Replicates).

Models	ACC	AUROC	AUPRC
LR	0.624 (0.620–0.628)	0.659 (0.645–0.673)	0.480 (0.471–0.489)
RF	0.635 (0.623–0.647)	0.670 (0.654–0.685)	0.464 (0.458–0.470)
LSTM	0.763 (0.762–0.764)	0.744 (0.743–0.745)	0.539 (0.537–0.542)
VHP	0.745 (0.743–0.747)	0.743 (0.741–0.744)	0.615 (0.612–0.617)
GRU	0.785 (0.783–0.788)	0.780 (0.775–0.786)	0.603 (0.601–0.605)
TLSTM	0.793 (0.793–0.794)	0.778 (0.772–0.784)	0.628 (0.624–0.632)
ATTAIN	0.810 (0.805–0.816)	0.782 (0.781–0.783)	0.640 (0.638–0.641)
FLTD	0.815 (0.813–0.816)	0.778 (0.775–0.780)	0.632 (0.628–0.637)
Transformer	0.810 (0.808–0.813)	0.780 (0.779–0.781)	0.646 (0.642–0.650)
StageNet	0.823 (0.822–0.824)	0.786 (0.783–0.789)	0.658 (0.656–0.661)
CTKGR	0.831 (0.831–0.832)	0.798 (0.797–0.799)	0.459 (0.458–0.460)

The performance results of the ablation study are presented in Table 5. As shown, the GRU model's performance in both ACC and AUROC metrics was significantly weaker compared to other variants. Further comparison between CTKGR-TLAM-Att and CTKGR-TLAM-TI revealed that the model utilizing only the attention mechanism (CTKGR-TLAM-TI) outperformed the model relying solely on time interval factors (CTKGR-TLAM-Att) in ACC and AUROC metrics. However, both CTKGR-TLAM-Att and CTKGR-TLAM-TI underperformed the complete CTKGR model in terms of the AUPRC metric.

3.4. Embedding visualization

We visualize the node embeddings of CTKGR and two baseline models, TA-DisMult and StageNet, to evaluate the effectiveness of feature representation learning. As shown in Fig. 2, the embeddings are generally clustered according to their entity types, indicating that all models can capture the intrinsic semantic features of different medical entities. Specifically, Fig. 2(a) illustrates the embeddings generated by our proposed CTKGR model, which exhibit a more complex and distinctive distribution pattern. This suggests that CTKGR can effectively model the differences and interactions among heterogeneous medical entities. In contrast, Fig. 2(b) and 2(c) show the embeddings learned by TA-DisMult and StageNet, respectively. Although the entity types remain separable, the overall distributions are simpler and less structured. Both TA-DisMult and StageNet can be regarded as homogeneous graph models, while CTKGR is a heterogeneous graph model. This comparison highlights that homogeneous GNNs tend to overlook the fine-grained semantic interactions between heterogeneous entities.

Overall, these visualization results demonstrate that CTKGR learns richer and more discriminative node representations, and more effectively captures the complex relational structures within the temporal pneumonia knowledge graph than the baseline models.

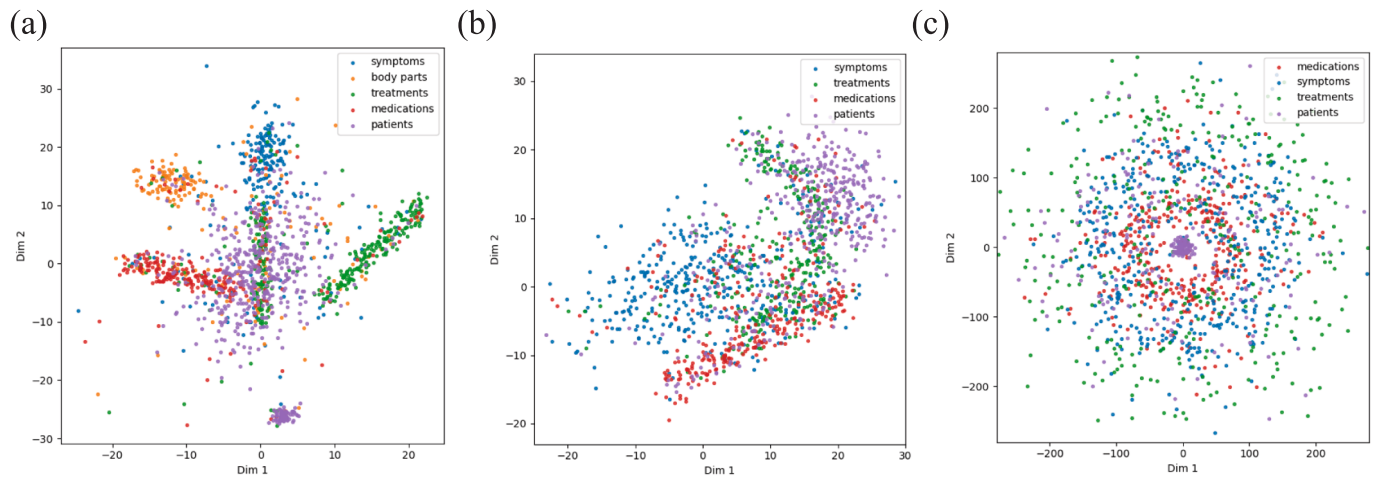


Fig. 2. T-SNE scatter plot of the embeddings. (a) CTKGR T-SNE plot; (b) TA-DisMult T-SNE plot; (c) StageNet T-SNE plot.

4. Discussion

4.1. Summary of model performance and advantages

The intertwined semi-structured data and temporal information in clinical EHRs present significant challenges in effectively representing patient data. This study introduces a graph-structured approach to clearly illustrate the evolution of medical entities and their relationships over time. Based on clinical records of pneumonia patients, we constructed a pneumonia TKG and developed the temporal graph model CTKGR to effectively represent its features.

Compared with traditional sequential models such as GRU and Transformer, CTKGR demonstrates significant advantages in various medical prediction tasks. Traditional GRU models and their variants primarily learn semantic information through sequential processing, but they often fail to capture deep interactions across different positions within sequences. Transformer models, despite using attention mechanisms to capture internal structural relationships within sequences, blend semantic and structural information into a single textual representation vector, lacking explicit structural guidance. In contrast, knowledge graphs explicitly represent associations between terms through nodes and edges. Thus, CTKGR directly optimizes the graph structure without relearning this structured information from the original text. Additionally, constructing the knowledge graph involves integrating and screening clinical information, reducing noise and highlighting key relationships. This provides the model with high-quality structured prior knowledge, significantly improving its comprehension of medical records.

The temporal distribution of clinical records is often irregular and contains missing data. Experiments showed that the GRU model's nearest-neighbor interpolation method for handling missing data yielded limited results [31]. This explicit interpolation separates the imputation process from the prediction task, making it difficult for the model to effectively capture underlying data patterns and potentially interfering with identifying low-frequency key features. By comparison, models like FLTD, StageNet, and CTKGR treat missing value imputation as joint vector embedding training, significantly enhancing model performance. These methods treat timestamps as events, incorporating additional time interval and inter-feature interaction information, effectively compensating for information loss due to ignored timestamps.

As the key module of CTKGR, the effectiveness of TLAM lies in its ability to model both the interactions among all temporal states and the intervals between events. As demonstrated by Deng et al., incorporating temporal intervals enables the model to control how much information should be inherited from previous time points, thereby addressing the problem of irregular time distributions. However, this information tends to decay as the time gap increases, potentially overlooking earlier but influential features.

Moreover, using only time intervals as features may assign equal weight to interpolated and real observations, neglecting their reliability differences. By introducing an attention mechanism, TLAM captures correlations among temporal features and dynamically adjusts the weights of inherited information, allowing more reliable interpolation for missing data. The combined effect of the attention mechanism and temporal interval modeling enables CTKGR to simultaneously capture the informational value of missingness and enhance the reliability of data imputation, thus improving overall model robustness.

4.2. Performance differences and analysis across tasks

In the knowledge graph construction, we report the performance of the BERT-CRF model on named entity recognition and relation extraction after each of the first seven annotation-training cycles. We added the result in Supplementary Fig. 2. With the increase in the number of training iterations, the size of the annotated corpus gradually expanded, and the model performance continuously improved.

In the link prediction task, CTKGR achieved the best performance on the Hits@1, Hits@3, and Hits@10 metrics in the ICEWS18 dataset, demonstrating superior ranking ability and local prediction accuracy. Although the xERTE model performed slightly better on the MRR metric due to its precise temporal relational attention mechanism, CTKGR's MRR was significantly superior to other baseline models, demonstrating greater robustness and generalization capabilities. In the pneumonia temporal knowledge graph constructed in this study, despite the relatively weaker overall performance of the TTransE model, it still showed some potential in the Hits@10 metric ~~(39.45%)~~. This suggests that providing some redundant candidate predictions in clinical settings could be beneficial for early risk warning and treatment recommendations.

In pneumonia clinical event prediction tasks, CTKGR generally achieved favorable results. However, relatively lower AUPRC values in symptom prediction and medication recommendation tasks indicated limitations when handling imbalanced datasets, possibly due to the model's tendency to focus on more abundant classes. Future research could incorporate re-weighted loss or contrastive sampling strategies to improve predictions for minority classes. Additionally, the GRU model, using nearest-neighbor clinical text for missing data imputation, failed to effectively capture temporal variations. While FLTD and StageNet improved performance by incorporating time deviation features, they did not sufficiently capture long-term interactions between time units. CTKGR's TLAM module effectively integrated attention mechanisms

with temporal intervals, significantly improving the reliability of missing data imputation and providing new methods for irregular temporal medical data modeling.

The ablation study further clarified the contributions of TLAM's components. Results demonstrated that implicit imputation methods outperformed explicit methods, with the attention-based CTKGR-TLAM-TI showing notable performance in ACC and AUROC metrics. However, both CTKGR-TLAM-Att and CTKGR-TLAM-TI still underperformed in the AUPRC metric, likely due to class imbalance issues. This indicates the necessity of further optimizing loss functions, sample weighting, or data augmentation techniques to enhance the model's handling of imbalanced data. We also did the error analysis by visualizing the patient node and the top-20 recommended drug nodes for two representative patients to analyze model performance (Supplementary Fig. 3).

4.3. Ontology extensibility

Our ontology was designed for modularity and extensibility, allowing new disease domains or healthcare systems to be incorporated by adding or modifying modular components (e.g., introducing new concepts or relationship types without disrupting the existing schema). Moreover, the ontology can be adapted to conditions with different clinical workflows by extending the relationship layer to capture time-dependent interactions between clinical entities. This design ensures that the framework is generalizable and can be readily adapted for broader clinical applications beyond pneumonia.

4.4. Clinical application prospects and future expansion

By integrating temporal knowledge graphs and graph representation learning techniques, CTKGR captures within-stay dynamics in pneumonia and translates them into actionable decision support. In real time, it delivers two outputs aligned with routine decision moments (pre-rounds, evening review) and sentinel events: a 24-h symptom-risk card (driving targeted reassessment/labs, respiratory-therapy scheduling, and nursing-plan adjustments) and a medication Top-k card with rationale (supporting antibiotic escalation, de-escalation, or maintenance). To ensure safe deployment, we could embed clinician-in-the-loop verification, prohibit autonomous order placement, and surface uncertainty indicators for high-variance predictions.

Additionally, the intuitive and interpretable nature of graph structures positions CTKGR as a valuable tool for medical education and clinical research, aiding medical students and clinicians in understanding disease progression. Future research could integrate multimodal data such as imaging and laboratory tests, further broadening application scope and enhancing prediction accuracy.

4.5. Limitations

Several limitations remain in this study. Firstly, due to the limited number of pneumonia cases, the knowledge graph constructed may lack comprehensive coverage and detailed information expression. Future work should expand dataset scales to enhance completeness and granularity. Secondly, CTKGR's performance on imbalanced tasks like symptom prediction and medication recommendation was moderate, suggesting room for improvement through loss function optimization or data augmentation strategies. Furthermore, despite significant enhancements in temporal data handling via the TLAM module, errors might still occur with extremely long intervals or consecutive missing data. Future studies could employ advanced interpolation techniques such as Bayesian interpolation or temporal variational autoencoders to further optimize model performance. Finally, we used CoreNLP in the initial annotation stage; nevertheless, its general-domain design may limit performance on clinical narratives, and future work will evaluate and integrate clinical NLP toolkits specifically developed for biomedical and EHR text.

This study was conducted on a single-center inpatient pneumonia cohort with a pneumonia-specific ontology; cross-disease, multi-center, and temporal generalizability were not evaluated, so any applicability beyond pneumonia should be interpreted as methodological potential rather than validated clinical generalizability. This study focuses on non-COVID inpatient pneumonia. We also excluded patients who died within admission because an observation window is required to define next-interval labels and their care goals differ fundamentally. This choice may introduce survivor bias and limits generalizability to ultra-early mortality or end-of-life pathways.

5. Conclusion

This study constructed a pneumonia TKG based on clinical records of pneumonia patients and proposed CTKGR method for dynamic modeling and prediction of clinical events. Experimental results demonstrate that CTKGR outperforms traditional sequential models across multiple medical tasks, including link prediction, symptom prediction, and medication recommendation. Additionally, the proposed TLAM effectively addresses issues of irregularity and missing data in clinical records. The method presented provides novel insights into predicting dynamic clinical events during hospitalization and has the potential to be extended to dynamic predictive tasks in other medical domains, offering new perspectives and effective support for clinical decision-making.

CRediT authorship contribution statement

Yingliang Yang: Software, Methodology. **Binyu Gao:** Methodology, Conceptualization. **Chunting Tan:** Data curation. **Congmin Zhu:** Data curation, Conceptualization. **Pixian Yuan:** Data curation. **Peng Xie:** Project administration, Methodology, Conceptualization. **Yu Gu:** Project administration, Methodology, Conceptualization. **Honglei Liu:** Supervision, Project administration, Conceptualization.

Funding

This work was supported by the Beijing Natural Science Foundation (No. 7242264, L246059), and the R&D Program of Beijing Municipal Education Commission (No. KM202310025020).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbi.2026.105007.

Data availability

Data will be made available on request.

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